

Metagenome-assembled genomes of acetogenic enrichment obtained from deep subsurface archean granitoids of Koyna Seismogenic Zone, Koyna, India

Rajendra Prasad Sahu,¹ Pinaki Sar¹

AUTHOR AFFILIATION See affiliation list on p. 4.

ABSTRACT We report 14 metagenome-assembled genomes (MAGs) of acetogenic bacteria (acetogens) enriched from deep (1,679–2,912 m below surface), hot (55°C–74°C) granitoids of the Koyna Seismogenic Zone, India. These MAGs include *Thermoanaerobacter pseudethanolicus*, *Exiguobacterium alkaliphilum*, *Moorella humiferrea*, *Caldanaerobacter subterraneus*, etc. The study allowed access to the genomes of deep biosphere acetogens.

KEYWORDS acetogenic bacteria, terrestrial deep biosphere, Koyna Seismogenic Zone, metagenome-assembled genome, archean granitoids

Acetogens are important members of energy-starved deep biosphere (1). Despite their critical ecological roles in driving primary production in deep biosphere through the Wood–Ljungdahl pathway (2) and biotechnological potential in C1 gas valorization (3), the genomes of these organisms remain understudied. Here, we present 14 bacterial MAGs enriched from deep, hot, Archean basement granitoids.

Seven rock cores spanning a depth range of 1,679–2,912 m and temperature gradient 50°C–74°C were sampled through 3,000 m deep Koyna pilot-borehole (Koyna seismogenic zone, Koyna, 17°17′57.27″ N, 73°44′19.07″ E), India. Standard procedures were followed to minimize samples' contamination (4, 5). Details of drilling, contamination-free rock sample collection, etc. were mentioned elsewhere (6, 7). Acetogens were enriched by adding rock powder into acetogenic medium (2 g/100 mL) in serum vial supplemented with H₂ + CO₂ (80:20 v/v) in the headspace (8). An A35 Don Whitely anaerobic workstation was used. Considering the *in situ* condition, three different incubation temperatures (50°C, 60°C, and 70°C) were used for the enrichment (details of the sampling depth, *in situ* and incubation temperatures of rock samples are provided as Table S1 at <https://doi.org/10.6084/m9.figshare.28215026>).

Following 6 months' incubation, 10 mL culture from each set was centrifuged (14,000×g, 30 min), cell pellet was used for total DNA extraction using QIAGEN Powersoil DNA isolation kit. Pair-end library was synthesized using Illumina DNA Prep, (M) Tagmentation kit followed by shotgun metagenome sequencing using Illumina NovaSeq 6000 (2 × 101 bp chemistry). Default parameters were used for sequencing and all software unless otherwise stated. Sequenced reads were demultiplexed (Illumina bcl2fastq [2.20]) and following adaptor removal (Skewer v0.2.2 [9]), their quality was analyzed (FastQC v0.12.1 [10]) and trimmed (Trimmomatic v0.39 [11]). Human contamination reads were removed (bowtie2 v2.5.3 together with samtools v1.7 and human host genome GRCh38 bowtie2 database [12, 13]). Quality filtered reads were assembled using MEGAHIT-1.2.9 (14), followed by binning through METABAT2 v1.2.9 (15), CONCOCT v1.1.0 (16), and MaxBin2 v2.2.7 (17). Resulting bins were consolidated and refined using Bin_refinement module (metaWRAP v1.3 [18]). GC%, completeness, and

Editor Frank J. Stewart, Montana State University, Bozeman, Montana, USA

Address correspondence to Pinaki Sar, sarpinaki@yahoo.com.

The authors declare no conflict of interest.

See the funding table on p. 5.

Received 25 August 2024

Accepted 30 January 2025

Published 25 February 2025

Copyright © 2025 Sahu and Sar. This is an open-access article distributed under the terms of the [Creative Commons Attribution 4.0 International license](https://creativecommons.org/licenses/by/4.0/).

TABLE 1 Detailed taxonomy, GeneBank accession numbers, and statistics of 14 metagenome-assembled genomes (MAGs) obtained from acetogenic enrichment of deep subsurface Archean granitoids present underneath the Deccan Traps, India

Mag id	Taxonomy (as per GTDBtk)	GeneBank accession	Completeness (%)	Contamination (%)	Genome coverage (x)	GC (%)	Size (Mb)	No. of contig	N50 (Mb)	Total CDS	RNA
AA2	Bacteria; Bacillota; Clostridia; Clostridiales; Clostridiaceae; MSJ-11	JBFNFA0000000000	100	1.61	177.34	30.3	4.06	33	0.27	3,990	43
^a AA7	Bacteria; Bacillota; Thermoanaerobacteria; Thermoanaerobacteraceae; Thermoanaerobacter; Thermoanaerobacter pseudethanolicus	JBFNEX0000000000	97.6	0	799.97	34.4	2.26	63	0.05	2,323	58
AA8	Bacteria; Bacillota; Bacilli; Exiguobacterales; Exiguobacteraceae; Exiguobacterium; Exiguobacterium alkaliphilum	JBFNEX0000000000	98.34	0	164.86	52.8	2.9	48	0.11	2,950	31
AA11	Bacteria; Pseudomonadota; Gammaproteobacteria; Enterobacterales; Enterobacteriaceae; Klebsiella; Klebsiella aerogenes	JBFNET0000000000	99.37	0.27	4.1	55.2	4.89	154	0.06	4,671	46
AA13	Bacteria; Bacillota; Bacilli; Bacillales; DSM-18226; Neobacillus	JBFNER0000000000	91.81	3.38	11.06	38.7	4.8	87	0.11	4,686	33
AA15	Bacteria; Pseudomonadota; Gammaproteobacteria; Burkholderiales; Burkholderiaceae; Burkholderia; Burkholderia contaminans	JBFNEP0000000000	100	0.42	20.16	66.2	8.66	83	0.17	7,845	59
^a AA17	Bacteria; Bacillota; Clostridia; Clostridiales; Clostridiaceae; Clostridium; Clostridium sp002453475	JBFNEN0000000000	98.46	1.9	8.6	30	3.67	65	0.09	3,451	37
AA18	Bacteria; Pseudomonadota; Gammaproteobacteria; Pseudomonadales; Pseudomonadaceae; Pseudomonas; Pseudomonas aeruginosa	JBFNEM0000000000	90.78	0.71	8	66.9	6.17	365	0.03	6,005	16
^a AA20	Bacteria; Bacillota; Thermoanaerobacteria; Thermoanaerobacteraceae; Thermoanaerobacter; Caldanaerobacter; Caldanaerobacter subterraneus	JBFNEK0000000000	97.2	1.13	37.85	37.7	2.27	139	0.02	2,341	41

(Continued on next page)

TABLE 1 Detailed taxonomy, GeneBank accession numbers, and statistics of 14 metagenome-assembled genomes (MAGs) obtained from acetogenic enrichment of deep subsurface Archean granitoids present underneath the Deccan Traps, India (Continued)

Mag id	Taxonomy (as per GTDBtk)	GeneBank accession	Completeness (%)	Contamination (%)	Genome coverage (x)	GC (%)	Size (Mb)	No. of contig	N50 (Mb)	Total CDS	RNA
AA21	Bacteria; Bacillota; Bacilli; Bacillales; Bacillaceae; <i>Bacillus anthracis</i>	JBFNEJ0000000000	98.61	3.86	13.63	35.1	5.28	34	0.28	5,447	21
AA22	Bacteria; Pseudomonadota; Gammaproteobacteria; Pseudomonadales; Pseudomonadaceae; <i>Stutzerimonas stutzeri</i>	JBFNEI0000000000	99.53	0.14	23.13	64.1	4.38	74	0.15	4,111	51
AA23	Bacteria; Bacillota; Symbiobacteriia; Symbiobacteriales; Symbiobacteriaceae; <i>Symbiobacterium</i>	JBFNEH0000000000	98.01	0	4.87	68.4	3.66	145	0.05	3,383	46
^a AA25	Bacteria; Bacillota; Moorellia; Moorelliales; Moorellaceae; <i>Moorella</i> ; <i>Moorella humiferrea</i>	JBFNEG0000000000	99.48	1.36	3.84	54	2.5	205	0.03	2,739	49
AA27	Bacteria; Bacillota; Bacilli; Bacillales; DSM-18226; <i>Cytobacillus cyrobacillus gotheilii</i>	JBFNEE0000000000	97.53	1.14	16.77	39.1	4.01	64	0.1	3,901	20

^aGenera reported as acetogens in AcetoBase2 (25).

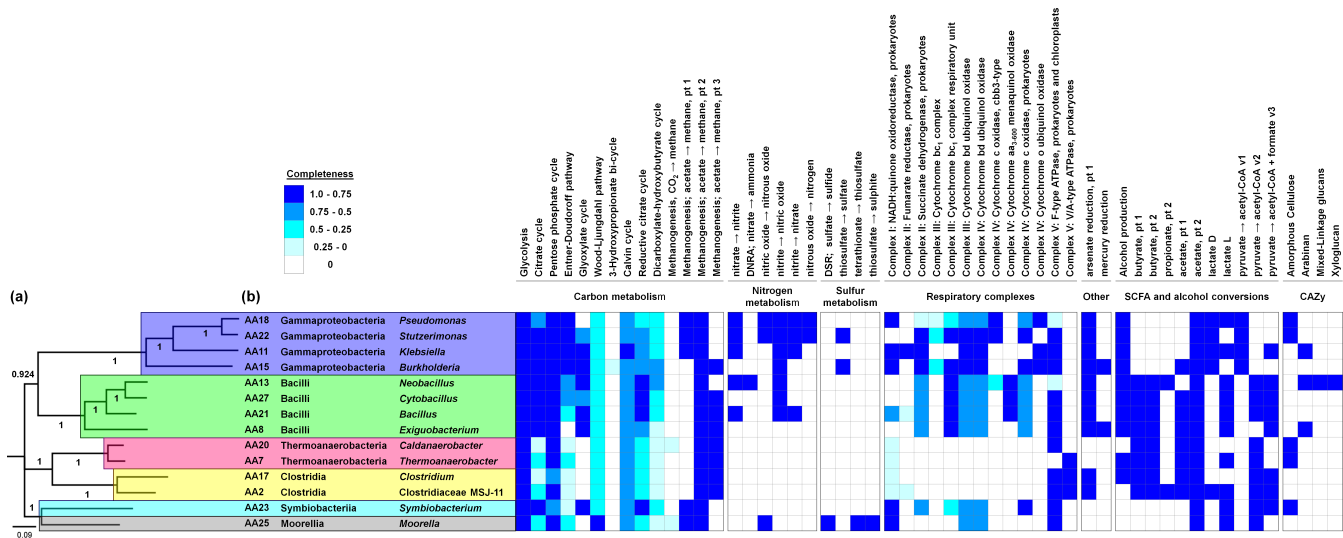


FIG 1 (a) Maximum likelihood tree of 14 metagenome-assembled genomes (MAGs). The phylogenetic tree was constructed using a concatenated set of 120 conserved bacterial marker genes from each of these MAGs using GTDB-Tk “classify” function and FigTree v1.4.4.. (b) Heatmap displays the pathways fully or partially present in each MAG, as predicted by kb_DRAM (24).

contamination percentage were determined (checkM v1.2.2 [19]), followed by genome coverage calculation (quant_bins module of metaWRAP v1.3 [18]). High-quality bins [>90% completeness, <5% contamination, total contigs < 500, N50 >20 kb [20, 21]] were taxonomically affiliated through GTDB-Tk v2.1.1 (22), and resulting alignments were used to build phylogenetic tree using FigTree v1.4.4 (<http://tree.bio.ed.ac.uk/software/figtree/>). Bins were annotated with PGAP v6.7 and kb_DRAM v0.1.2 (23, 24).

During this study 113.62 million quality filtered reads were assembled into 29,442 contigs. These were binned into 14 high-quality MAGs with average size and GC content 4.25 Mb and 48.06%, respectively (Table 1). MAGs consisted of 33–365 contigs (average N50 value 108.57 kb) with coverage varied between 3.84x and 799.97x. Annotation of MAGs showed presence of 2,323–7,845 number of coding DNA sequences (CDS). Taxonomic diversity and functional potential of the MAGs are presented in Fig. 1.

These MAGs provide access to the genomes of deep rocky-biosphere acetogens allowing us to gain insights into their significance in functioning and adaptation of deep life.

ACKNOWLEDGMENTS

Granitoid rock core samples used in this study were collected during drilling of Koyna pilot borehole (KFD1) supported by Ministry of Earth Sciences (MoES), Govt. of India. Authors gratefully acknowledge the support from MoES-Borehole Geophysics Research Laboratory, Karad, India, particularly, Dr Sukanta Roy for their constant support. Funding support from MoES vide Project ID: MoES/P.O.(Seismo)/1(288)/2016 dated 16 March 2017 is gratefully acknowledged. R.P.S. gratefully acknowledges his fellowship provided by Council of Scientific and Industrial Research, Government of India.

AUTHOR AFFILIATION

¹Department of Bioscience and Biotechnology, Indian Institute of Technology Kharagpur, Kharagpur, West Bengal, India

AUTHOR ORCIDs

Rajendra Prasad Sahu  <http://orcid.org/0000-0001-8740-6780>
Pinaki Sar  <http://orcid.org/0000-0001-8538-2527>

FUNDING

Funder	Grant(s)	Author(s)
Ministry of Earth Sciences, India	MoES/P.O.(Seismo)/1(288)/2016	Pinaki Sar

AUTHOR CONTRIBUTIONS

Rajendra Prasad Sahu, Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review and editing | Pinaki Sar, Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review and editing

DATA AVAILABILITY

Raw sequence reads are available in the NCBI GenBank database under Sequence Read Archive accession numbers [SRX24612229](https://www.ncbi.nlm.nih.gov/sra/SRX24612229) (BioProject [PRJNA1021524](https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1021524)). Draft genomes are available under Genbank accession numbers [PRJNA1121667](https://www.ncbi.nlm.nih.gov/genbank/PRJNA1121667).

REFERENCES

- Beaver RC, Neufeld JD. 2024. Microbial ecology of the deep terrestrial subsurface. *ISME J* 18:wrae091. <https://doi.org/10.1093/ismejo/wrae091>
- Momper L, Casar CP, Osburn MR. 2023. A metagenomic view of novel microbial and metabolic diversity found within the deep terrestrial biosphere at DeMMO: A microbial observatory in South Dakota, USA. *Environ Microbiol* 25:3719–3737. <https://doi.org/10.1111/1462-2920.16543>
- Bae J, Park C, Jung H, Jin S, Cho B-K. 2024. Harnessing acetogenic bacteria for one-carbon valorization toward sustainable chemical production. *RSC Chem Biol* 5:812–832. <https://doi.org/10.1039/d4cb00099d>
- Colwell FS, Boyd S, Delwiche ME, Reed DW, Phelps TJ, Newby DT. 2008. Estimates of biogenic methane production rates in deep marine sediments at hydrate ridge, cascadia margin. *Appl Environ Microbiol* 74:3444–3452. <https://doi.org/10.1128/AEM.02114-07>
- Kieft TL, Phelps TJ, Fredrickson JK. 2007. Drilling, coring, and sampling subsurface environments. *Man Environ Microbiol*:799–817.
- Sahu RP, Kazy SK, Bose H, Mandal S, Dutta A, Saha A, Roy S, Dutta Gupta S, Mukherjee A, Sar P. 2022. Microbial diversity and function in crystalline basement beneath the deccan traps explored in a 3 km borehole at Koyna, western India. *Environ Microbiol* 24:2837–2853. <https://doi.org/10.1111/1462-2920.15867>
- Roy S. 2017. Scientific drilling in Koyna region, Maharashtra. *Curr Sci* 112:2181.
- Hallbeck L, Pedersen K. 2008. Characterization of microbial processes in deep aquifers of the fennoscandian shield. *Appl Geochem* 23:1796–1819. <https://doi.org/10.1016/j.apgeochem.2008.02.012>
- Jiang H, Lei R, Ding SW, Zhu S. 2014. Skewer: a fast and accurate adapter trimmer for next-generation sequencing paired-end reads. *BMC Bioinformatics* 15:1–12. <https://doi.org/10.1186/1471-2105-15-182>
- Andrews S. 2010. FastQC: a quality control tool for high throughput sequence data. Cambridge, United Kingdom.
- Bolger AM, Lohse M, Usadel B. 2014. Trimmomatic: a flexible trimmer for Illumina sequence data. *Bioinformatics* 30:2114–2120. <https://doi.org/10.1093/bioinformatics/btu170>
- Langmead B, Salzberg SL. 2012. Fast gapped-read alignment with Bowtie 2. *Nat Methods* 9:357–359. <https://doi.org/10.1038/nmeth.1923>
- Li H, Handsaker B, Wysoker A, Fennell T, Ruan J, Homer N, Marth G, Abecasis G, Durbin R. 2009. Subgroup 1000 genome project data processing. *bioinformatics* 25:2078–2079. <https://doi.org/10.1093/bioinformatics/btp352>
- Li D, Liu C-M, Luo R, Sadakane K, Lam T-W. 2015. MEGAHIT: an ultra-fast single-node solution for large and complex metagenomics assembly via succinct de Bruijn graph. *Bioinformatics* 31:1674–1676. <https://doi.org/10.1093/bioinformatics/btv033>
- Kang DD, Li F, Kirton E, Thomas A, Egan R, An H, Wang Z. 2019. MetaBAT 2: an adaptive binning algorithm for robust and efficient genome reconstruction from metagenome assemblies. *PeerJ* 7:e7359. <https://doi.org/10.7717/peerj.7359>
- Alneberg J, Bjarnason BS, de Bruijn I, Schirmer M, Quick J, Ijaz UZ, Lahti L, Loman NJ, Andersson AF, Quince C. 2014. Binning metagenomic contigs by coverage and composition. *Nat Methods* 11:1144–1146. <https://doi.org/10.1038/nmeth.3103>
- Wu Y-W, Simmons BA, Singer SW. 2016. MaxBin 2.0: an automated binning algorithm to recover genomes from multiple metagenomic datasets. *Bioinformatics* 32:605–607. <https://doi.org/10.1093/bioinformatics/btv638>
- Uritskiy GV, DiRuggiero J, Taylor J. 2018. MetaWRAP—a flexible pipeline for genome-resolved metagenomic data analysis. *Microbiome* 6:158. <https://doi.org/10.1186/s40168-018-0541-1>
- Parks DH, Imelfort M, Skennerton CT, Hugenholtz P, Tyson GW. 2015. CheckM: assessing the quality of microbial genomes recovered from isolates, single cells, and metagenomes. *Genome Res* 25:1043–1055. <https://doi.org/10.1101/gr.186072.114>
- Bowers RM, Kyrpides NC, Stephanoukas R, Harmon-Smith M, Doud D, Reddy TBK, Schulz F, Jarett J, Rivers AR, Elie-Fadrosh EA, et al. 2017. Minimum information about a single amplified genome (MISAG) and a metagenome-assembled genome (MIMAG) of bacteria and archaea. *Nat Biotechnol* 35:725–731. <https://doi.org/10.1038/nbt.3893>
- Meziti A, Rodriguez-R LM, Hatt JK, Peña-Gonzalez A, Levy K, Konstantinidis KT. 2021. The reliability of metagenome-assembled genomes (MAGs) in representing natural populations: insights from comparing MAGs against isolate genomes derived from the same fecal sample. *Appl Environ Microbiol* 87:e02593. <https://doi.org/10.1128/AEM.02593-20>
- Chaumeil P-A, Mussig AJ, Hugenholtz P, Parks DH. 2020. GTDB-Tk: a toolkit to classify genomes with the genome taxonomy database. Oxford University Press. <https://doi.org/10.1093/bioinformatics/btz848>
- Li W, O'Neill KR, Haft DH, DiCuccio M, Chetvernin V, Badretdin A, Coulouris G, Chitsaz F, Derbyshire MK, Durkin AS, Gonzales NR, Gwadz M, Lanczycki CJ, Song JS, Thanki N, Wang J, Yamashita RA, Yang M, Zheng C, Marchler-Bauer A, Thibaud-Nissen F. 2021. RefSeq: expanding the prokaryotic genome annotation pipeline reach with protein family model curation. *Nucleic Acids Res* 49:D1020–D1028. <https://doi.org/10.1093/nar/gkaa1105>
- Shaffer M, Borton MA, Bolduc B, Faria JP, Flynn RM, Ghadermazi P, Edirisinghe JN, Wood-Charlson EM, Miller CS, Chan SHJ, Sullivan MB, Henry CS, Wrighton KC. 2023. kb_DRAM: annotation and metabolic profiling of genomes with DRAM in KBase. *Bioinformatics* 39:btad110. <https://doi.org/10.1093/bioinformatics/btad110>
- Singh A, Schnürer A. 2022. AcetoBase version 2: a database update and re-analysis of formyltetrahydrofolate synthetase amplicon sequencing data from anaerobic digesters. *Database (Oxford)* 2022:baac041. <https://doi.org/10.1093/database/baac041>